bode_analysis.sce

```
clear;
clc;

m = 1823; Iz = 3500; a = 1.42; b = 1.45; Cf = 120000; Cr = 180000; V = 27.78;

A = [0, 1, 0, 0;
     0, -(2*Cf + 2*Cr)/(m*V), 0, -V - (2*Cf*a - 2*Cr*b)/(m*V);
     0, 0, 0, 1;
     0, -(2*Cf*a - 2*Cr*b)/(Iz*V), 0, -(2*Cf*a^2 + 2*Cr*b^2)/(Iz*V)];
B = [0; (2*Cf)/m; 0; (2*Cf*a)/Iz];
C_y = [1, 0, 0, 0];
D_y = 0;

sys_vehicule = syslin('c', A, B, C_y, D_y);

Kp = 0.1800; Ki = 0.0100; Kd = 0.0500;
s = poly(0, 's');
G_vehicule = ss2tf(sys_vehicule);
C_pid = Kp + Ki/s + Kd*s;

sys_FTBO = G_vehicule * C_pid;
sys_FTBO = clean(sys_FTBO);
sys_FTBO = syslin('c', sys_FTBO);

[g_marge, f_marge_g, p_marge, f_marge_p] = margin(sys_FTBO);

printf("Marge de Gain  : %.2f dB à %.2f Hz\n", g_marge, f_marge_g);
printf("Marge de Phase : %.2f deg à %.2f Hz\n", p_marge, f_ma
```

```
rge_p);

scf(8); clf();
f_min = 0.001; f_max = 1000;
bode(sys_FTBO, f_min, f_max);
show_margins(sys_FTBO);
xtitle("Figure 8 : Analyse fréquentielle (Bode) et Marges de
stabilité du système LKA");
```